Time Frequency Analysis

Effect of Window

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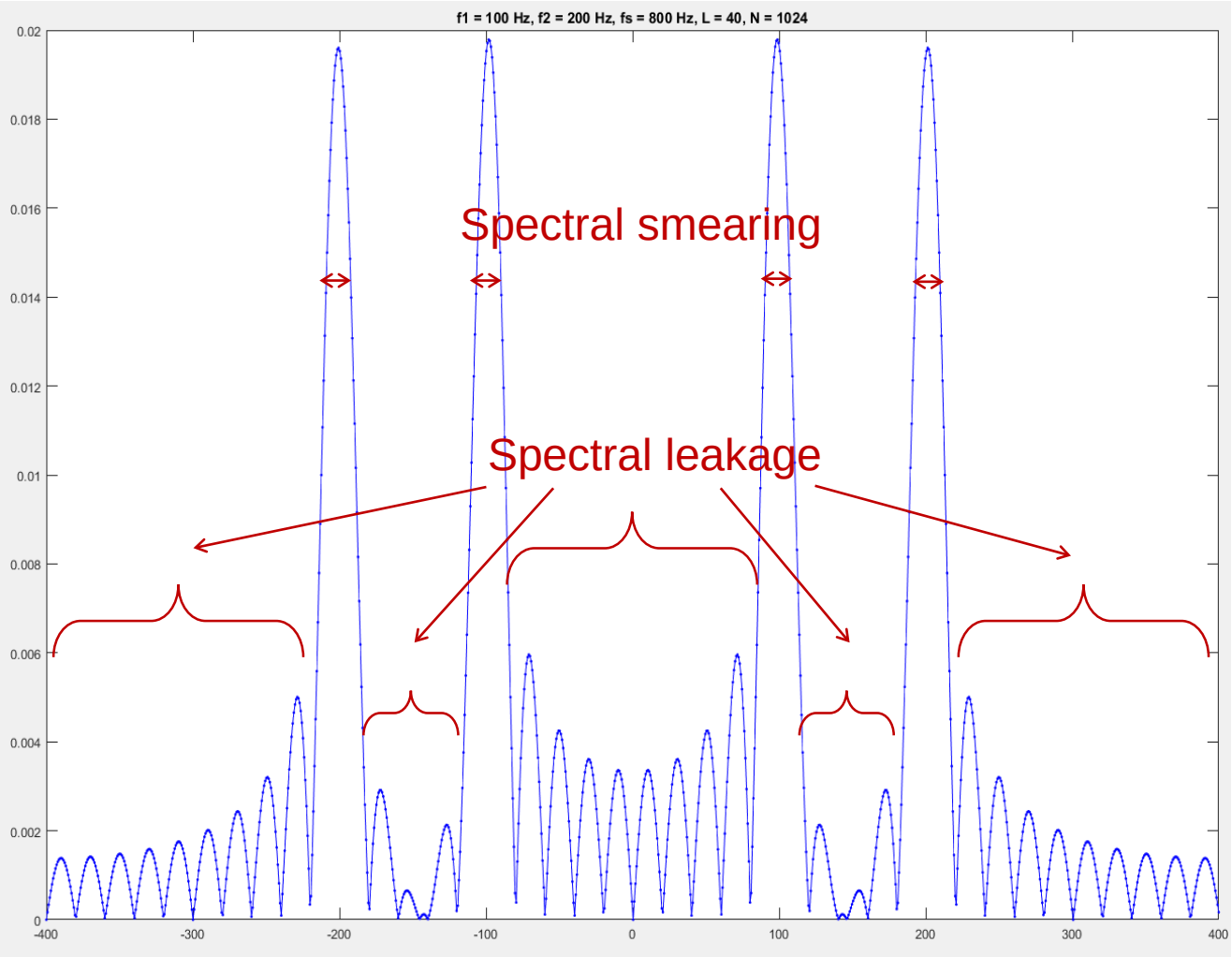
Example

- $y = \sin(2\pi f_1 t) + \sin(2\pi f_2 t)$
- f_s = 800Hz
- Rectangular window of size *L*
- *N*-pt FFT

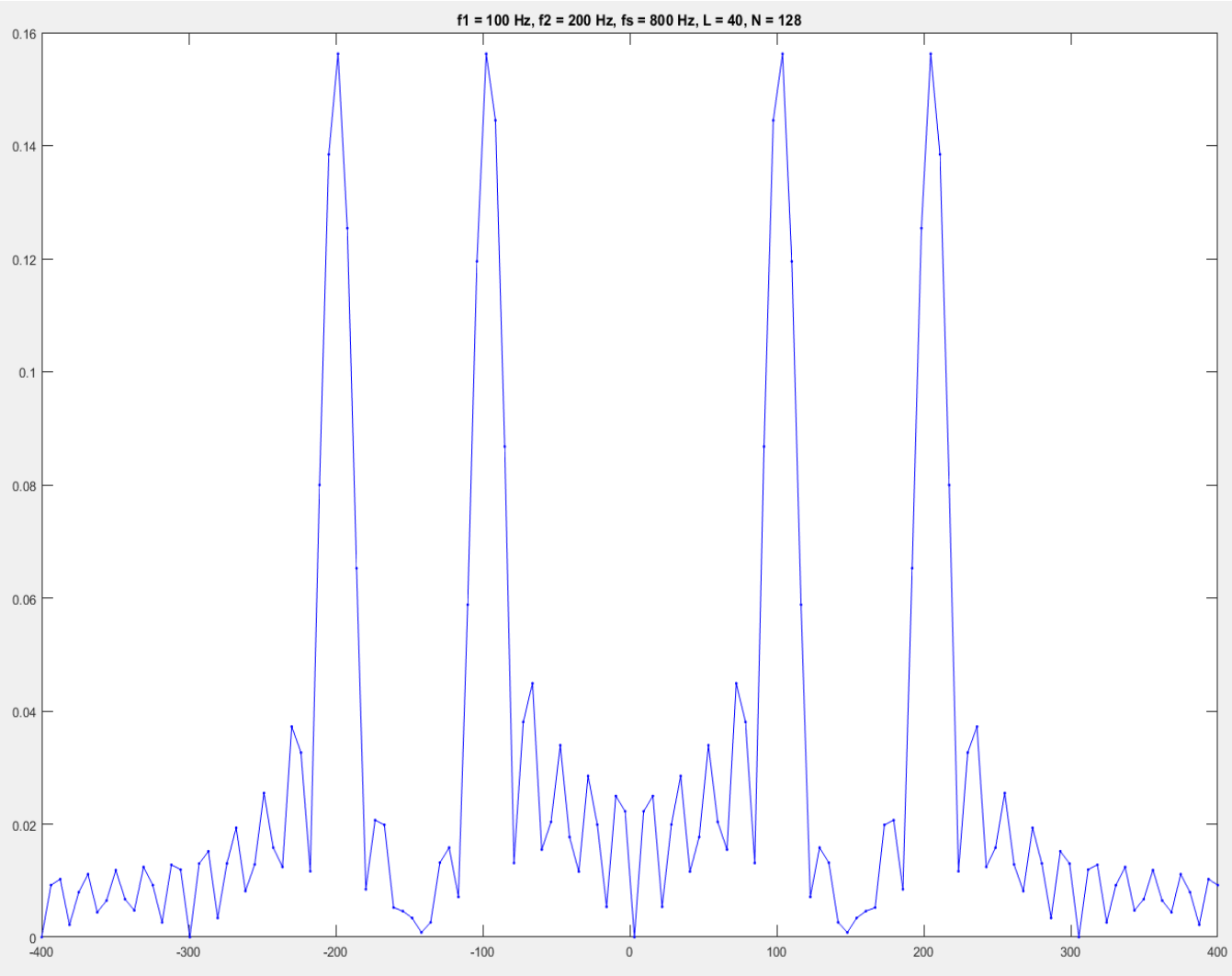
Effect of Window

- f_1 = 100Hz
- f_2 = 200Hz
- L = 40
- N = 1024



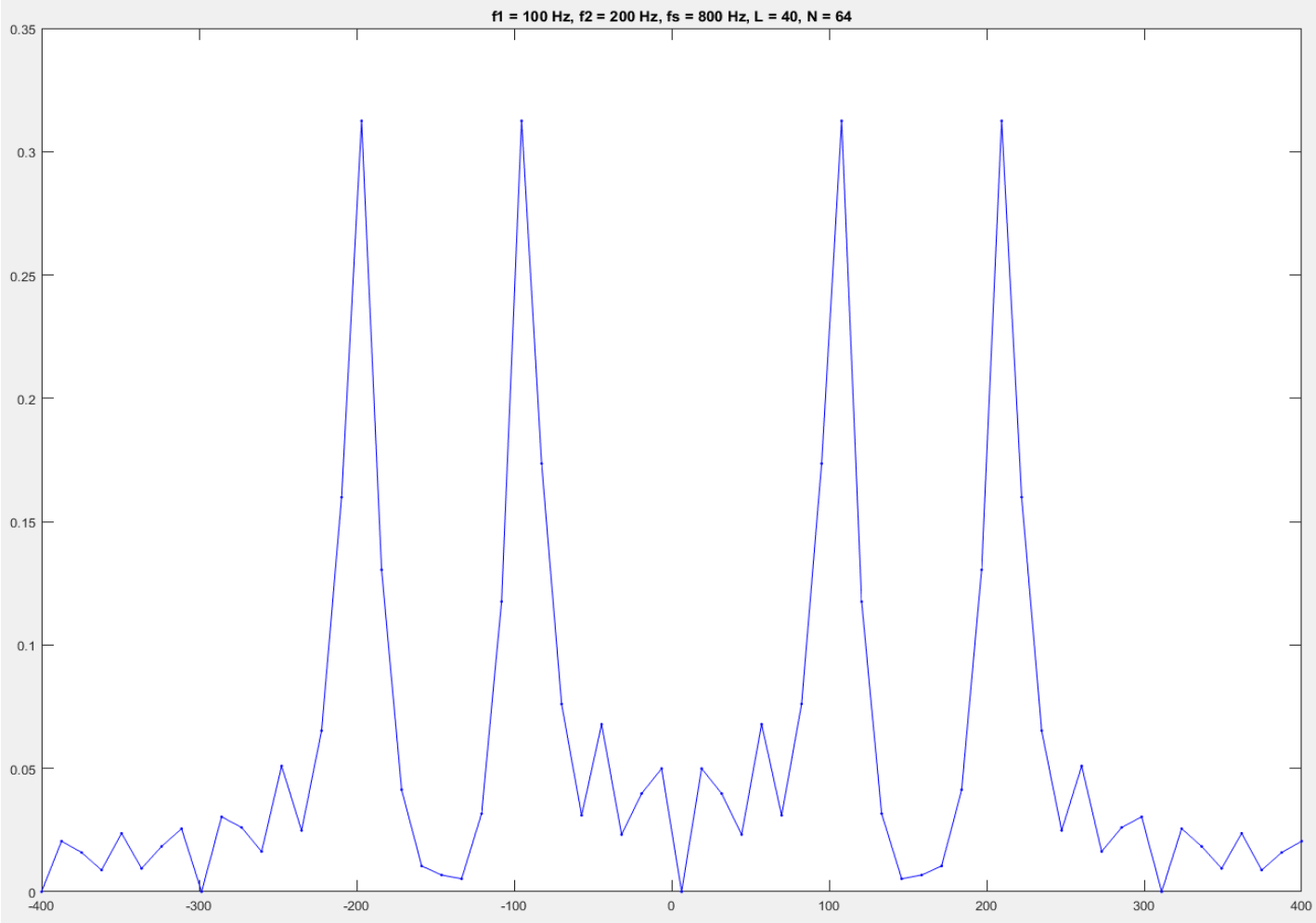
Effect of Window

- f_1 = 100Hz
- f_2 = 200Hz
- L = 40
- Change frequency resolution:
  - N = 128



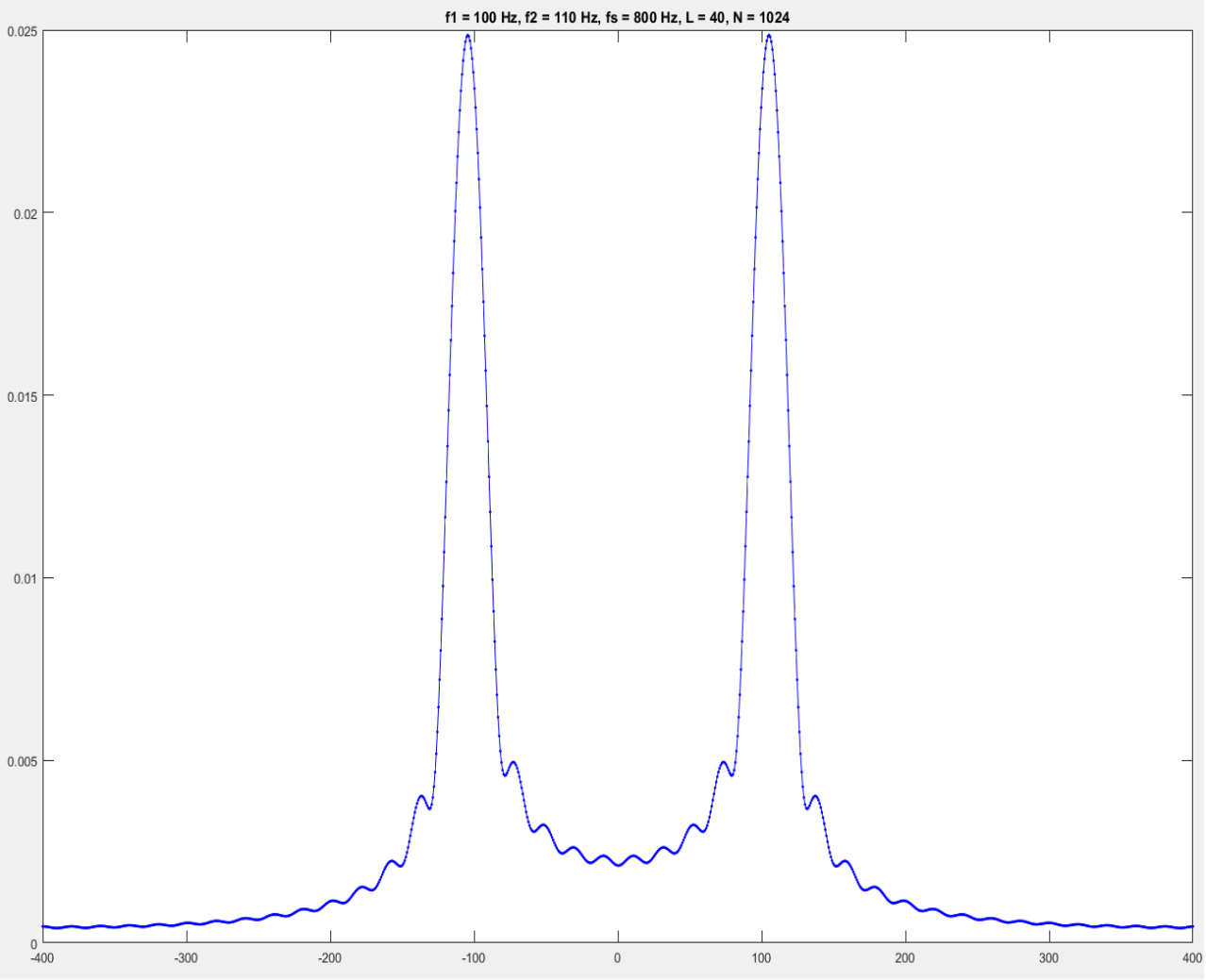
Effect of Window

- f_1 = 100Hz
- f_2 = 200Hz
- L = 40
- Change frequency resolution:
  - N = 64



Effect of Window

- f_1 = 100Hz
- f_2 = **110Hz**
- L = 40
- N = 1024
- f_1 and f_2 can not be accurately identified!



Trade-offs

- Different window types: main lobe width vs side lobe height, need to provide adequate frequency selectivity.
- As *L* (window size) increases, main lobe width decreases, frequency resolution increases, time resolution decreases, and vice versa.
- As *N* (FFT size, $N \geq L$) increases, frequency resolution improves, but computation complexity also increases.
- Frequency resolution will impact if an event (peak) in frequency domain will be identified accurately or not.
- Time resolution will identify when the event occurred.

Question

Which of the following statements is NOT correct?

a) Larger FFT size demands more computation.

b) Smaller window size gives better time resolution.

c) Smaller window size gives better frequency resolution.

d) There is a tradeoff between main lobe width and side lobe height of different window functions.

Exercise

Consider the following discrete-time signal *x[n]* that is obtained by sampling the analog signal with a sampling frequency of 1Hz:

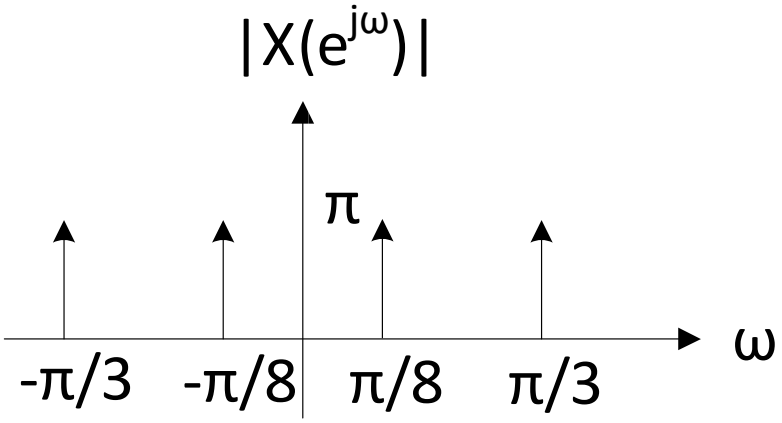
$$x[n] = sin\left(\frac{\pi}{3}n\right) + sin\left(\frac{\pi}{8}n\right)$$

- Determine and plot the magnitude of the DTFT $|X(e^{j\omega})|$ of the signal *x[n]*.
- Let *w[n]* be a causal rectangular window of length *L*. The window *w[n]* is used to extract *L*-point segments from the signal *x[n]*. Determine the minimum window length *L* that can be used to resolve the frequency components of *x[n]*. Let *N* be the minimum 2^{α} number that is greater than or equal to *L*. Determine the frequency resolution of the *N*-point DFT.
- Repeat the last question with a causal Hamming window of length *L'*.

Answer

DTFT:

$$X\left(e^{j\omega}\right) = \frac{\pi}{j}\delta\left(\omega - \frac{\pi}{3}\right) - \frac{\pi}{j}\delta\left(\omega + \frac{\pi}{3}\right) + \frac{\pi}{j}\delta\left(\omega - \frac{\pi}{8}\right) - \frac{\pi}{j}\delta(\omega + \frac{\pi}{8})$$



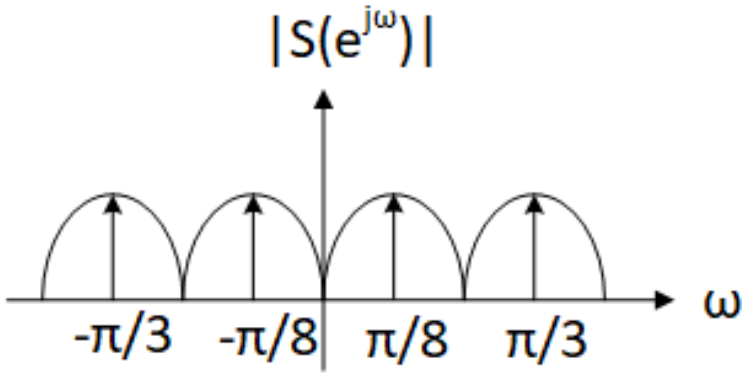
Answer

- $s[n] = x[n]w[n] \overset{DTFT}{\longleftrightarrow} S(e^{j\omega}) = \frac{1}{2\pi} X(e^{j\omega}) * W(e^{j\omega})$

- $S(e^{j\omega})$

$$= \frac{1}{2\pi} W(e^{j\omega}) * \left(\frac{\pi}{j}\delta\left(\omega - \frac{\pi}{3}\right) - \frac{\pi}{j}\delta\left(\omega + \frac{\pi}{3}\right) + \frac{\pi}{j}\delta\left(\omega - \frac{\pi}{8}\right) - \frac{\pi}{j}\delta\left(\omega + \frac{\pi}{8}\right) \right)$$

$$= \frac{1}{2j}\left(W\left(e^{j\left(\omega - \frac{\pi}{3}\right)}\right) - W\left(e^{j\left(\omega + \frac{\pi}{3}\right)}\right) + W\left(e^{j\left(\omega - \frac{\pi}{8}\right)}\right) - W\left(e^{j\left(\omega + \frac{\pi}{8}\right)}\right)\right)$$



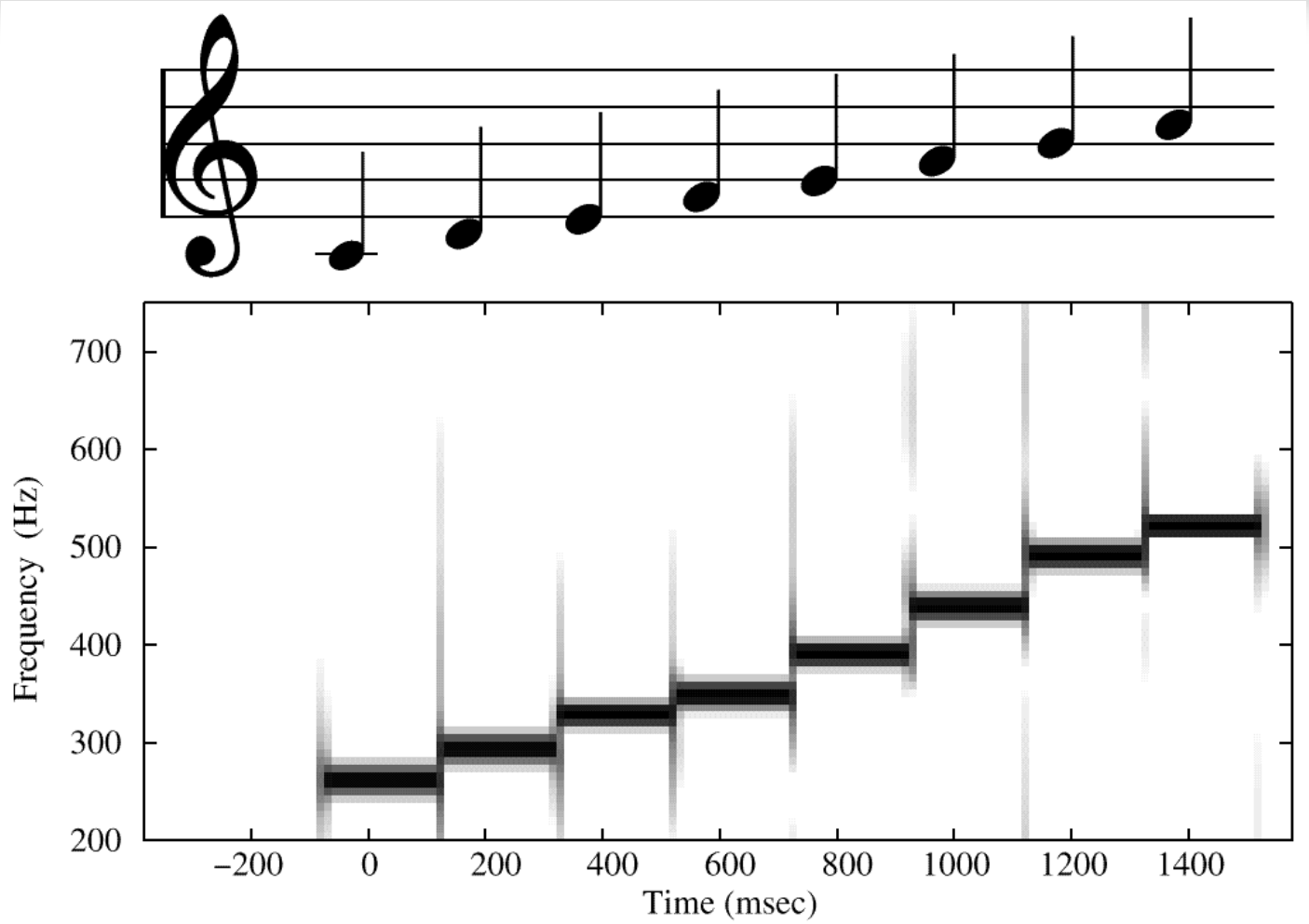
Answer

- **To resolve the frequency components of *x[n]*, the width of the main lobe of the window function needs to be smaller than the minimum frequency spacing of the frequency components of *x[n]*, otherwise main lobes overlap, which could potentially generate wrong peaks.**

- **For a rectangular window,**
  - The main lobe width $4\pi/L \leq \pi/3 - \pi/8$, i.e., $L \geq 96/5 = 19.2 \Rightarrow L \geq 20$
  - $N = 2^{\alpha} = 32 \geq L \geq 20$
  - Frequency resolution $= 2\pi/N = \pi/16$ radians or $\pi/16 * f_s/(2\pi) = 1/32$ Hz

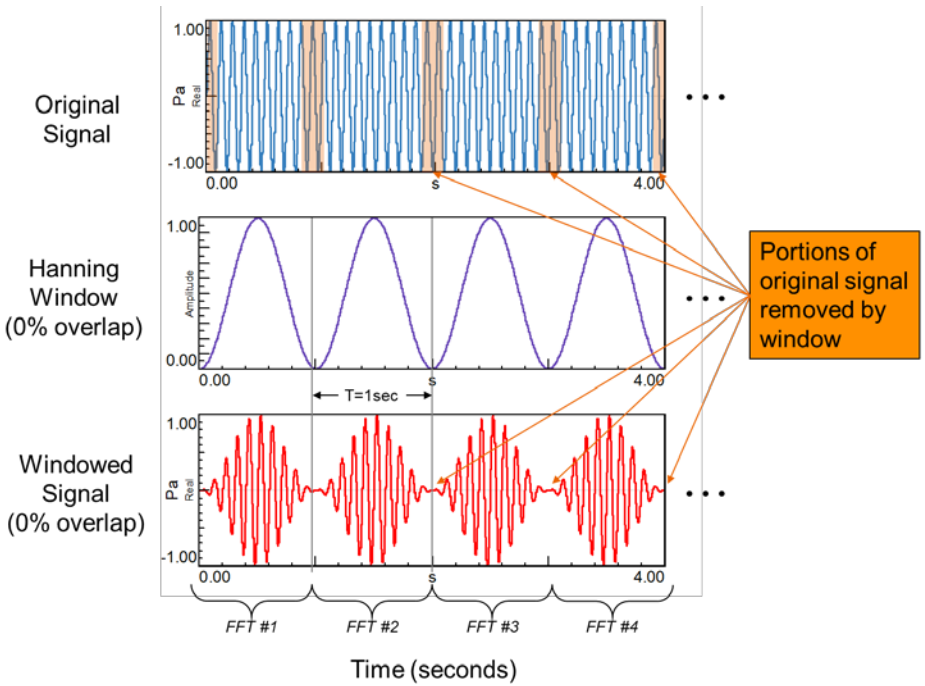
- **For a Hamming window,**
  - The main lobe width $8\pi/L \leq \pi/3 - \pi/8$, i.e., $L \geq 192/5 = 38.4 \Rightarrow L \geq 39$
  - $N = 2^{\alpha} = 64 \geq L \geq 39$
  - Frequency resolution $= 2\pi/N = \pi/32$ radians or $\pi/32 * f_s/(2\pi) = 1/64$ Hz

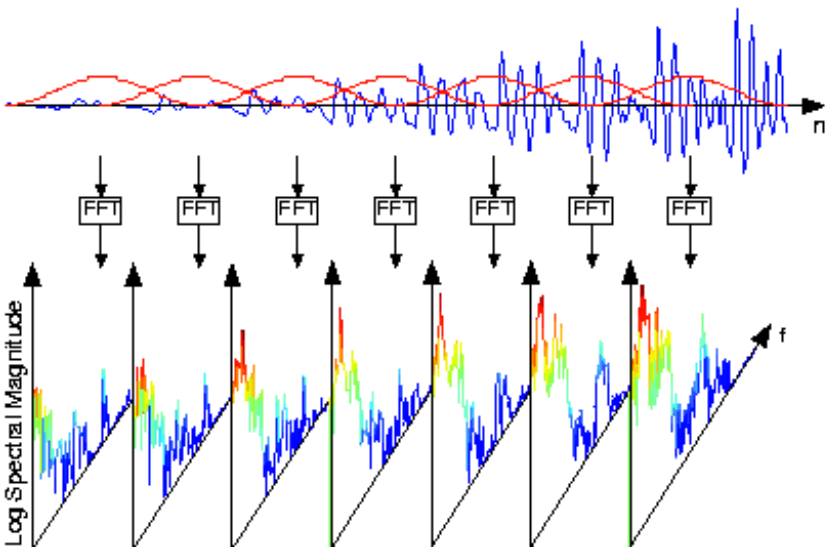
Recall: C-Major Scale Spectrogram Example



Output from spectrogram, analysis frame (200 ms), artifacts at transitions.

Overlapping Windows





Hanning window with 50% overlap

Images from: https://community.sw.siemens.com/s/article/Overlap-What-Why-and-How-to-use-it
https://eng.libretexts.org/Bookshelves/Electrical_Engineering/Introductory_Electrical_Engineering/Electrical_Engineering_(Johnson)/05%3A_Digital_Signal_Processing/5.10%3A_Spectrograms

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